1. Study guide structure

CAA01 — Foundations of Information Systems

4 chapters

Chapters 1, 2, 3 and 4

The foundations of information systems, organisational strategy, ethics, privacy and information technology infrastructure.

CAA02 — Data, Networks and Business Applications

4 chapters

Chapters 5, 7, 9 and 10

Networking, e-business, organisational information systems and customer-oriented applications.

CAA03 — Managing and Developing Information Systems

4 chapters

Chapters 11, 12, 13 and 14

Information systems management, business intelligence, systems acquisition and development, and emerging technologies, depending on the edition's chapter structure.

One important consideration: we should verify the exact chapter titles of your ninth-edition International Adaptation, because titles and numbering can differ between editions. That will prevent you from studying material outside your UOC syllabus.

2. Chapter-by-chapter summaries

I have now located the publisher's chapter list, so we can follow the correct 2022 edition.



Introduction to Information Systems
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These are original, condensed revision notes aligned with the textbook's chapter topics, rather than extracts from the complete textbook.

CAA01 — Foundations of Information Systems

Chapter 1. Introduction to Information Systems

Main idea: Information systems combine people, processes and technology to collect, process, store and distribute information.

Key definitions

- Data: Raw facts without interpretation, such as individual sales transactions.
- Information: Data processed into a meaningful form.
- Knowledge: Understanding obtained by interpreting information and applying experience.
- Information System (IS): A combination of people, processes, data and technology that supports organisational activities.
- Information Technology (IT): The technological resources used by an information system, including hardware, software and networks.

Important concepts: Computer-based information systems, business processes, operational efficiency, decision-making and the impact of IT on organisations and society.

Example: A supermarket records individual purchases (data), calculates daily sales (information), and identifies which products to reorder (knowledge applied to a decision).

Remember: IT is part of an information system, but an information system also includes people and organisational procedures.

Chapter 2. Organizational Strategy, Competitive Advantage, and Information Systems

Main idea: Organisations use information systems to improve business processes and achieve competitive advantages.

Key definitions

- Business process: A sequence of activities that produces a particular business outcome.
- Competitive advantage: A characteristic that allows an organisation to outperform its competitors.
- Business Process Improvement (BPI): Incremental improvements to existing processes.
- Business Process Reengineering (BPR): Fundamental redesign of business processes to achieve major improvements.
- Business Process Management (BPM): Continuous management, analysis and optimisation of business processes.

Important concepts: Porter's Five Forces, value chains, cost leadership, differentiation, innovation and strategic alignment between business objectives and IT.

Example: A logistics company uses route-optimisation software to reduce delivery times and operational costs.

Remember: BPI improves an existing process, BPR radically redesigns it, and BPM manages ongoing process performance.

Chapter 3. Ethics and Privacy

Main idea: Organisations must use information systems responsibly and protect personal information.

Key definitions

- Information ethics: Moral principles governing the use of information and technology.
- Privacy: The ability of individuals to control access to and use of their personal information.
- Intellectual property: Legally protected creations, including software and digital content.
- Informed consent: Agreement given after receiving understandable information about how personal data will be used.
- Digital divide: Inequality in access to or effective use of digital technologies.

Important concepts: Data protection, employee monitoring, surveillance, copyright, ethical decision-making and the GDPR in the European Union.

Example: An online store must establish a lawful basis for processing customers' personal data and explain relevant processing activities.

Remember: A technological action can be legally permitted but still raise ethical concerns.

Chapter 4. Information Security

Main idea: Information security protects information systems against threats, attacks and unauthorised access.

Key definitions

- Confidentiality: Only authorised parties can access information.
- Integrity: Information remains accurate and protected against unauthorised modification.
- Availability: Information and systems remain accessible when needed.
- Authentication: Verification of a user's identity.
- Authorisation: Determination of what an authenticated user is permitted to do.
- Encryption: Transformation of information into a form unreadable without the appropriate decryption mechanism.

Important concepts: Malware, phishing, ransomware, vulnerabilities, risk assessment, backups, access controls, firewalls and multi-factor authentication.

Example: A banking application uses MFA to reduce account takeover risk, encryption to protect information, and backups to support recovery.

Remember: The CIA triad — Confidentiality, Integrity and Availability — is fundamental to information security.

CAA02 — Data, E-Commerce and Organisational Systems

Chapter 5. Data and Knowledge Management

Main idea: Organisations need to organise, maintain and analyse data to support decisions.

Key definitions

- Database: An organised collection of related data.
- Database Management System (DBMS): Software used to create, maintain and access databases.
- Relational database: A database that organises information into related tables.
- Primary key: An attribute or combination of attributes uniquely identifying a record.
- Foreign key: An attribute referencing a key in another table.
- Data warehouse: A repository integrating data, commonly for reporting and analytics.
- Knowledge management: Processes for capturing, sharing and applying organisational knowledge.

Important concepts: Structured and unstructured data, SQL, database relationships, data quality, Big Data, data governance and knowledge sharing.

Example: A company stores customer orders in a relational database and analyses historical purchases through a data warehouse.

Remember: Operational databases primarily support transactions, whereas data warehouses are designed primarily for analysis.

Chapter 7. E-Business and E-Commerce

Main idea: Digital networks enable organisations to conduct transactions and business activities electronically.

Key definitions

- E-business: Use of digital technologies to support business activities and processes.
- E-commerce: Buying and selling products or services through electronic networks.
- B2B: Business-to-business transactions.

- B2C: Business-to-consumer transactions.
- C2C: Consumer-to-consumer transactions.
- Digital marketplace: An online platform connecting buyers and sellers.

Important concepts: Online payment systems, electronic storefronts, digital marketing, electronic procurement, platform business models and online transaction security.

Example: An online retailer receives an order, processes payment, updates inventory and arranges delivery using interconnected systems.

Remember: E-commerce is a subset of e-business. Not every digital business process involves an online sale.

Chapter 9. Social Computing

Main idea: Social technologies enable communication, collaboration and information sharing among individuals and organisations.

Key definitions

- Social computing: Technology-supported interaction and collaboration between users.
- Social network: A platform or structure connecting individuals and organisations.
- User-generated content (UGC): Content created and shared by platform users.
- Crowdsourcing: Obtaining ideas, information or services from a large group of participants.
- Social commerce: Commercial activities influenced or enabled by social platforms.
- Social CRM: Use of social technologies to support customer relationships.

Important concepts: Online communities, customer engagement, reputation management, social media analytics, collaborative content and privacy.

Example: A company monitors customer feedback on social media to identify recurring product problems.

Remember: Social computing can improve collaboration and customer engagement, but also introduces privacy, misinformation and reputation risks.

Chapter 10. Information Systems Within the Organization

Main idea: Organisations use different types of information systems to coordinate daily activities and support management decisions.

Key definitions

- Transaction Processing System (TPS): Records routine business transactions.

- Management Information System (MIS): Produces structured reports for management.
- Decision Support System (DSS): Supports analysis and decision-making, particularly for less structured problems.
- Enterprise Resource Planning (ERP): Integrates core business functions through shared processes and data.
- Functional Area Information System: Supports activities within a particular department.

Important concepts: Accounting systems, finance, human resources, production, inventory, reporting, organisational integration and decision support.

Example: An ERP system such as Odoo integrates sales, inventory, purchasing and accounting.

Remember: TPS focuses on recording transactions; MIS on reporting; DSS on analysis and decisions; ERP on integrating organisational processes.

CAA03 — Enterprise Management, Analytics and Emerging Technologies

Chapter 11. Customer Relationship Management and Supply Chain Management

Main idea: Organisations use information systems to manage relationships with customers, suppliers and business partners.

Key definitions

- Customer Relationship Management (CRM): Strategies, processes and technologies used to manage customer relationships.
- Supply Chain Management (SCM): Coordination of material, information and financial flows across a supply chain.
- Customer lifetime value (CLV): Estimated economic value generated by a customer relationship over time.
- Supply chain: Network of organisations involved in producing and delivering goods or services.
- Bullwhip effect: Amplification of demand fluctuations as information moves upstream through a supply chain.

Important concepts: Customer acquisition, retention, personalisation, sales automation, inventory management, logistics and supplier coordination.

Example: A CRM identifies customers likely to stop purchasing, while an SCM system helps ensure products are replenished efficiently.

Remember: CRM primarily focuses on customers and relationships; SCM focuses on coordinating supply and delivery networks.

Chapter 12. Business Analytics

Main idea: Business analytics converts data into insights that improve organisational decisions.

Key definitions

- Business Intelligence (BI): Technologies and processes that transform data into useful business information.
- Business analytics: Application of analytical techniques to investigate business performance and support decisions.
- Descriptive analytics: Examines what happened.
- Predictive analytics: Estimates what may happen.
- Prescriptive analytics: Recommends actions based on objectives and constraints.
- Key Performance Indicator (KPI): A measurable indicator of progress towards an objective.
- Dashboard: A visual interface presenting important metrics and indicators.

Important concepts: Data visualisation, statistical analysis, forecasting, data mining, reporting and data-driven decision-making.

Example: A retailer analyses historical sales, predicts future demand and recommends inventory quantities.

Remember: Descriptive = past performance; predictive = likely outcomes; prescriptive = recommended actions.

Chapter 13. Acquiring Information Systems and Applications

Main idea: Organisations must select, acquire, develop and maintain information systems that meet their requirements.

Key definitions

- Systems Development Life Cycle (SDLC): A structured approach to planning, analysing, designing, implementing and maintaining systems.
- Feasibility study: Evaluation of whether a proposed solution is viable.
- Requirements analysis: Identification of the needs and constraints a system must satisfy.
- Outsourcing: Contracting external organisations to perform services or development work.

- Software as a Service (SaaS): Software delivered through a service provider, usually accessed over a network.
- Agile development: An iterative approach emphasising collaboration, adaptation and incremental delivery.

Important concepts: Build-versus-buy decisions, software acquisition, project management, testing, implementation, change management and maintenance.

Example: A company compares purchasing an existing CRM with developing a custom application.

Remember: A successful system must satisfy business requirements, not merely function correctly from a technical perspective.

Chapter 14. Artificial Intelligence

Main idea: Artificial intelligence enables computer systems to perform tasks involving learning, reasoning, perception and decision support.

Key definitions

- Artificial Intelligence (AI): The field concerned with building systems that perform tasks associated with intelligent behaviour.
- Machine Learning (ML): Techniques that enable models to learn patterns from data.
- Deep Learning: Machine learning using multilayer neural networks.
- Natural Language Processing (NLP): Techniques for processing and generating human language.
- Expert system: A system that applies encoded knowledge and rules to problems in a specific domain.
- Robotic Process Automation (RPA): Software automation of repetitive, rule-based tasks; it does not necessarily require AI.

Important concepts: Recommendation systems, predictive models, intelligent automation, computer vision, chatbots, AI bias, explainability and responsible AI.

Example: An online retailer uses machine learning to recommend products based on purchasing patterns.

Remember: AI is the broader field; ML is one approach within AI, and deep learning is a subset of ML.

3. Quick revision: essential comparisons

Concepts	Main difference
Data vs. information	Raw facts vs. meaningful processed facts
IS vs. IT	Complete sociotechnical system vs. technological

	components
BPI vs. BPR	Incremental improvement vs. radical redesign
Authentication vs. authorisation	Who you are vs. what you may access
Database vs. data warehouse	Transactional data management vs. analytical storage
E-business vs. e-commerce	Digital business activities vs. online commercial transactions
CRM vs. SCM	Customer relationships vs. supply chain coordination
ERP vs. CRM	Enterprise-wide integration vs. customer-focused management
Predictive vs. prescriptive analytics	Forecasting outcomes vs. recommending actions
AI vs. ML	Broad intelligent-systems field vs. learning from data